

Their Performance

In our Tier 1 tests, the cards based on ATi Radeon 9550 chipset suffered badly in *Doom 3*. Although they did post playable frame-rates at lower resolutions, the price they retail at does not justify the performance. In fact, *Doom 3* was the overall stress test for all the cards we tested. It brought almost all the cards in the low category to their knees including the 9600s and the 5700s. The 5600 chip from last year was missing this time and was replaced by a new and improved 5700. The clock speeds have been cranked up as have the memory speeds with the Ultra's showing the most improvement. Similarly, the 9600XT is the improvised version of the 9600PRO with more memory and increased clock and memory speeds.

We tested all the cards for the Tier 1 tests and then filtered out four from each category to run the Tier 1 tests on. The criterion we decided to apply was performance. Better the frame-rates, higher the chances of it going into the next tier. In our Tier-1 tests, the Gigabyte duo and the PowerColor and XFX cards posted some of the best results.

In the tests, we kept the in-game settings across all resolutions at medium. At this medium setting, the PowerColor 9600XT was on fire and gave excellent frame-rates on all games beating the 5700 Ultra, which has a higher memory speed and also has a faster memory type. The



cards that made it to the next level were the PowerColor 9600XT, the XFX 5700 Ultra and the Gigabyte 5700 and 5700 LE.

In Tier 2, we cranked up the resolutions and also the in-game settings. All settings were set to 'High' and the minimum resolution we started from was 1024 x 768. This is where the XFX GeForce FX 5700 Ultra shone bright. With its higher memory speeds, and optimised pixel pipelines which handled two textures per pixel, it posted the best frame rates in Tier 2.

The PowerColor Radeon 9600 just went limp here; in fact, it crashed twice with visible artefacts in the higher resolutions posting a measly frame-rate of approximately 5 fps at 1280 x 1024 in *Far Cry*. On the other hand, the card did well with 4X AA ON, at 1024 x 768 in both *Half-Life 2* and *Far Cry*. But *Doom 3* was simply not its strength.

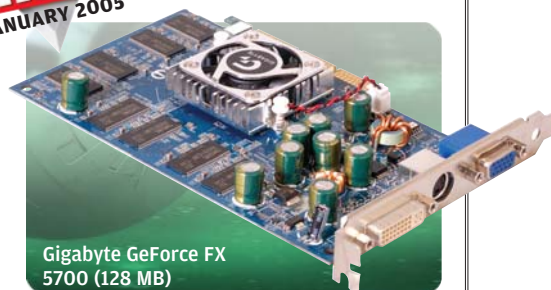
The XFX GeForce FX 5700 Ultra was just waiting to be turned on at the higher settings. The ceiling for this card was

similar to others in *Doom 3* and with 4X AA at 1024 x 768, cranking up the resolution any higher spelt 'doom' for the card and the game was simply unplayable. However, the 5700 Ultra provided very good frame rates for all other games we tested and can be considered for future games.

The Gigabyte 5700 was the second in line to lay claim for the throne in this category and posted good frame rates. With the competition heating up, it was difficult for us to look for the runner-up to the prized Digit Best Buy award.

The Winners Are...

One thing you need to keep in mind is that the cards we tested in this category are only good enough to play



games up to a resolution of 1024 x 768. You may get playable frame rates at higher frame rates with high in-game settings in some games, but if you don't, don't be surprised. You can rest assured, however, that these cards can deal with all games at higher resolutions if the in-game eye candy is kept low.

The Tier 2 test results immediately reflected the first winner, which was the XFX 5700 Ultra 256 MB, because of the frame-rates it posted, it was awarded with the Digit Best Buy Gold Award.

The low performance of the PowerColor 9600XT at 1280 x 1024 resolution was definitely puzzling and we ran the tests not once, but four times to double check our results. Nevertheless, the results remained the same. However, the value for money that the PowerColor Radeon 9600XT offers is unmatched by any other card in this category.

If you are low on money but still want a decent card, this card will not disappoint. The card next in line was the Gigabyte GeForce FX 5700 128 MB, which ended second and was awarded the Digit Best Buy Silver Award.

? Is PCI Express x16 Important?

PCI Express is the new technology that will invade consumer desktops starting this year.

Jumping up from 4X AGP to 8X AGP did not provide any major performance leap in terms of frame-rates in games. For the current crop of games that is played on desktops worldwide, AGP 4X is more than enough to provide the required bandwidth. However, the limiting factor is the memory management, which bottlenecks the data transfer. The increased bandwidth offered by AGP 8X does little to improve this. Card manufacturers have gone and improved the clock and memory speeds for the accelerator, but the overall system performance is not in their hands. Therefore, if you have a slower CPU and a faster card, games will still lag.

With PCI-e x16, the bandwidth has doubled from that offered by AGP 8X; however, the core issue still remains. Right now, any game

that runs comfortably on an AGP card will do the same on an x16 card; however, the PCI-Express solution offers better data transfer rates for other components on the motherboard. The main reason companies are adopting PCI-Express is because there is a huge performance gain that can be tapped in all areas of data transfer from one component to another.

As always, moving on from AGP to PCI-e will be more of a transition brought about by market conditions than an option exercised by the end-user. As soon as the current motherboards start disappearing from the market and PCI-e becomes prevalent, you will have to go for a PCI-e solution. If you have an AMD 64 or Intel 3.2 GHz and above system with a good 4X or 8X AGP card, our suggestion would be to sit tight and not upgrade, since PCI-e will definitely not make *Doom 3* run at 200fps at 1,600 x 1,200!